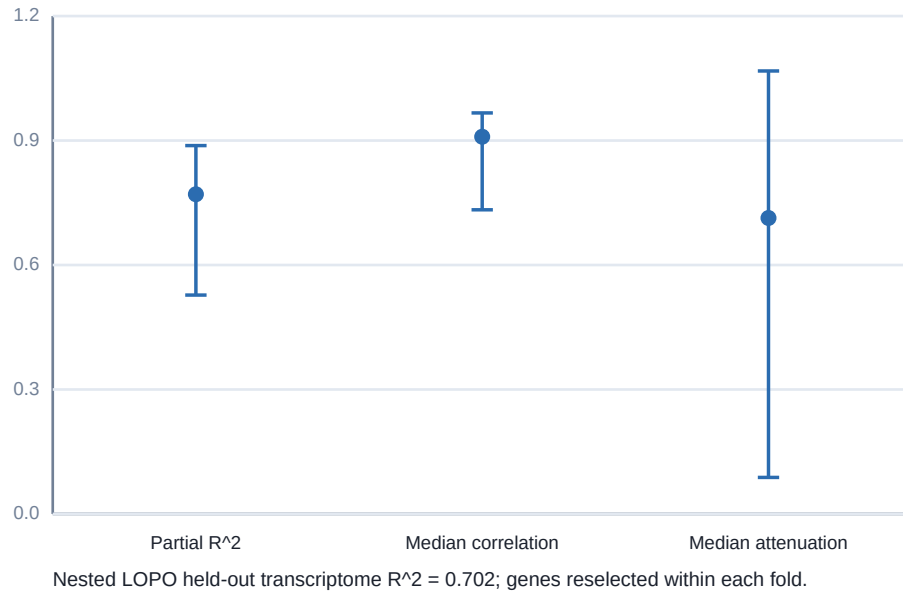
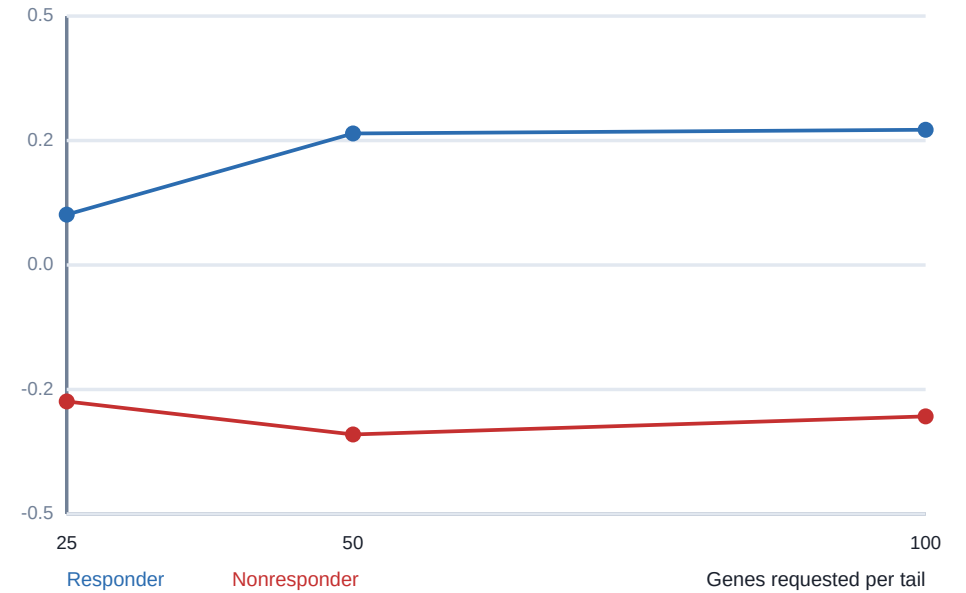


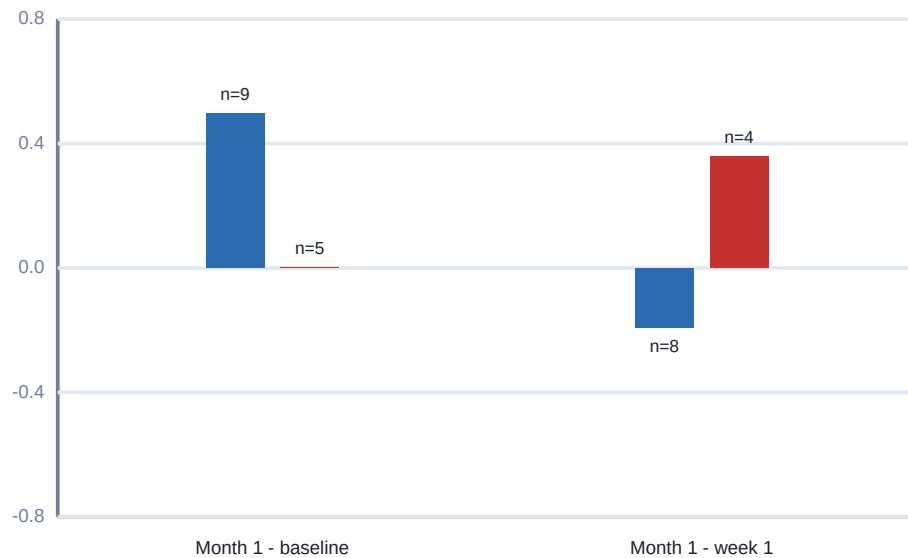
### A Variance attribution and patient-bootstrap uncertainty



### B Young-platelet residual sensitivity



### C Complete-case month-1 platelet-score changes



### D Operational composition-dominance framework

- 1 External lineage score
- 2 Within-patient coupling
- 3 Effect attenuation and partial  $R^2$
- 4 Lineage and matched-set controls
- 5 Patient-exclusion stability
- 6 Multiplicity-corrected residual test

*A bulk response is termed composition-dominant only when all six checks are concordant.*